PAPER_678: UQFF LISA vs LIGO Cross-Sensitivity Comparison and Crossover Frequency

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Domain: Gravitational Wave Detectors

Abstract

We compute the UQFF suppression ratio $R_{\text{supp}}(f) = h_{\text{UQFF}}(f)/h_{\text{GR}}(f)$ across both detector bands. A crossover frequency f_{cross} exists where LISA-band and LIGO-band suppressions are equal: dominated by $S_{\text{UA,LISA}}$ at low f and by S_{U_m} at high f. We show that UQFF corrections are $\sim 10\%$ larger in the LISA band than in the LIGO band for stellar-mass BH sources.

Primary UQFF Equation

$$R_{\text{supp}}(f) = (1 - f_{\text{TRZ}}) \cdot S_{\text{SCm}} \cdot \min(S_{U_m}, S_{\text{UA,LISA}})$$

Parameters

Band	Freq Range	Dominant UQFF Effect
LIGO	10–2000 Hz	S_{U_m}
LISA	10^{-4}–1 Hz	$S_{\text{UA,LISA}}$

UQFF Constants

Constant	Value	Description
ρ_{UA}	7.09×10^{-36} kg/m³	Universal Aether density
ρ_{SCm}	7.09×10^{-37} kg/m³	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor
κ	5×10^{-4} day⁻¹	UQFF calibration constant
$[\text{SSq}]$	0.57	Superstring quenching factor
μ_J	3.38×10^{23} J·m	Magnetic string coupling
γ	5×10^{-5} / 86400 s⁻¹	Aether oscillation decay

C++ Implementation

```cpp
#include "LISAVsLIGOComparisons.h"

LISAVsLIGOComparisons obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```python
from CondensedPhysics2 import LISAVsLIGOComparisonsCalculator

calc = LISAVsLIGOComparisonsCalculator()
result = calc.compute()  # returns all UQFF quantities
print(result)
```

References

LIGO O3 sensitivity 2021; Amaro-Seoane 2017 LISA; UQFF Session 173

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